

Securing the Digital Mine

A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in Industrial Mining IoT

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Track: Track 3: Smart Subsoil - Digital Transformation and Automation in the Mineral Resources Complex

Public Repository: github.com/mhiskall282/Securing-the-Digital-Mine-UNESCO-Project

75.61% Feature Pruning
41 telemetry attributes reduced to 10 via constrained BWOA
0.76 ms Edge Latency
207x faster than baseline on 1GB RAM Raspberry Pi 4B
0.82 MB Footprint
83.2% compression with post-training Float16 quantization

***SPEAKER NOTES:** Good morning session chairs and esteemed colleagues. Today our team from UEW Ghana presents our DSR artifact 'Securing the Digital Mine' under the UNESCO Smart Subsoil track.*

The Industrial Mining Threat Landscape

Air-Gap Erosion and Kinetic Cyber-Physical Risks in Mineral Processing

Modern Mineral Concessions: Gold (Ghana, South Africa) and strategic base-metal complexes (Russian Urals, Siberia) deploy dense IIoT telemetry.

The Air-Gap Myth: Cloud analytics, remote vendor maintenance tunnels, and centralized digital twins have eliminated physical network isolation.

Legacy Protocol Insecurity: Modbus RTU/TCP, DNP3, and OPC-UA transmit data in clear plaintext without cryptographic authentication or sequence integrity.

Kinetic Cyber Risks in Extraction Facilities:

- Semi-Autogenous Grinding (SAG) Mills (15 MW motors):* Overriding cooling valve setpoints causes catastrophic motor seizure (\$25k-\$50k/hr downtime).
- Tailings Storage Facilities (TSF):* Falsifying piezometer pressure data conceals dam slope instability, risking dam overtopping and toxic chemical spills.
- Underground Ventilation:* Tampering with variable-frequency drive (VFD) fans risks fatal gas accumulation and miner asphyxiation.

SPEAKER NOTES: In traditional enterprise IT, cybersecurity focuses on data confidentiality. In mining, physical safety and continuous availability dictate everything. A cyber attack modifying setpoints produces physical disaster.

AIC Triad in Mining
Availability & Integrity precede Confidentiality to prevent worker harm
Financial Exposure
Unplanned mining downtime costs \$50,000 to \$500,000 per hour
Kinetic Damage
Direct physical damage to heavy industrial milling & dewatering assets

The SCADA Real-Time Control Loop Dilemma

Why Conventional IT Deep Learning Fails in Industrial Extraction Environments

The Industrial Timing Mismatch:

- PLC Cyclic Scan Loop Deadline: **20 ms to 50 ms** (Strict control timing margin)
- Unoptimized Deep Learning IDS (41 features): **157.66 ms** (VIOLATES DEADLINE - causes buffer bloat)
- Proposed BWOA Quantized Model (10 features): **0.76 ms** (PASSES DEADLINE - 131x below ceiling)

Four Core Industrial Deficiencies in Existing Tools:

- Signature Brittleness:** Snort/Suricata evaluate byte strings. Attackers manipulating valid Modbus function codes (FC 05) bypass them (<15% recall).
- High Dimensionality:** Anomaly models trained on 41 to 80+ features overwhelm low-power ARM industrial gateway CPUs.
- Latency Bloat:** 157 ms processing latency drops incoming telemetry packets during high-speed SCADA polling loops.
- African Concession Realities:** Remote sites operate under solar microgrids, satellite backhaul, and 1GB RAM hardware.

50 ms Deadline
Industrial controllers execute scan loops every 20-50 ms
157.66 ms Baseline
Unoptimized neural networks drop packets and violate safety loops
0.76 ms Our Framework
Processes 1,300+ flows per second on a single low-power ARM core

SPEAKER NOTES: Industrial controllers scan sensors and actuate valves every 20 to 50 milliseconds. A security model requiring 157 milliseconds causes packet loss and safety violations. We engineered a 0.76 ms solution.

Research Questions Guiding the DSR Artifact

Methodological Formulation under the Design Science Research Paradigm

Grounding in Design Science Research (DSR): Peffers et al. (2007) and Hevner et al. (2004) framework.

Four Core Research Questions Addressed in this Investigation:

- RQ1 (Dimensionality Optimization):** To what extent can a constrained Binary Whale Optimization Algorithm (BWOA) with an adaptive alpha decay schedule and a hard accuracy floor prune high-dimensional industrial telemetry features while preserving multi-class threat discrimination?
- RQ2 (Spatial-Temporal Threat Modeling):** How effectively does a hybrid 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture capture packet-level spatial correlations and sequential connection state transitions in industrial SCADA networks?
- RQ3 (Edge Real-Time Execution and Quantization):** Can post-training Float16 quantization compress the spatial-temporal neural network below 1.0 MB and achieve sub-millisecond (<1.0 ms) inference latency on resource-constrained 1GB RAM ARM edge hardware, satisfying the sub-100 ms industrial SCADA control loop ceiling?
- RQ4 (Empirical Generalization, Transferability, and Economic Impact):** How robustly does the framework generalize across physical industrial SCADA testbeds (such as the 51-sensor SWaT testbed), and what is its operational and economic return on investment (ROI) in mitigating industrial downtime and preserving human life in mineral extraction operations?

SPEAKER NOTES: To ensure academic and engineering rigor, we formulated four explicit research questions. These guide our feature selection, neural architecture design, edge quantization, and socio-economic evaluation.

DSR Methodology
6 systematic stages from problem identification to communication
Empirical Grounding
Every RQ mapped to concrete statistical and hardware benchmarks
Holistic Scope
Covers algorithm, neural architecture, edge execution, and business ROI

End-to-End System Architecture: 4-Tier Edge Boundary

Decoupled Edge-Native Pipeline for Real-Time Threat Ingestion and Autonomous Mitigation

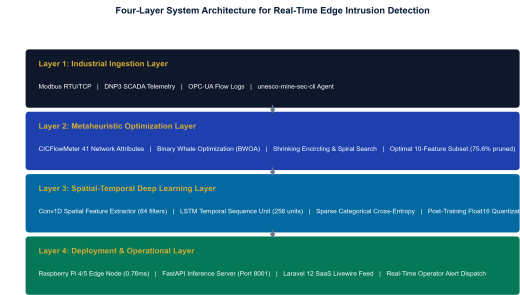
Tier 1 (Industrial Ingestion Layer): Non-blocking packet sniffer built with Node.js 20 and libpcap. Binds to switch mirror (SPAN) ports, extracting network flow vectors at line speed without inline delay.

Tier 2 (Metaheuristic Optimization Layer): Prunes incoming flow vectors using the 10 BWOA-selected features, discarding 75.61% of uninformative attributes in <0.05 ms.

Tier 3 (Deep Learning Classification Layer): Compiled TensorFlow Lite Float16 spatial-temporal engine. Executes thread-safe single-sample inference in 0.76 ms on ARM edge cores.

Tier 4 (Supervisory Visualization Layer): High-concurrency FastAPI microservice (port 8001) streaming predictions, class probabilities, and latency gauges to an industrial Livewire dashboard.

Offline Autonomy Guarantee: The entire 4-tier pipeline runs 100% locally on edge hardware with zero external cloud dependencies, maintaining defense during complete satellite backhaul severance.



Zero Cloud Dependency

Edge gateways defend operations even during complete satellite loss

Non-Blocking Sniffing

SPAN mirror ingestion avoids introducing network packet delay

Sub-Millisecond Pipeline

Total end-to-end ingestion and classification in under 1.0 ms

SPEAKER NOTES: Our architecture operates in four decoupled layers from packet capture to dashboard visualization. Because it runs completely on-premises on the gateway, mining sites are protected even when remote uplinks fail.

Constrained Binary Whale Optimization (BWOA)

Mathematical Formulation of Dimensionality Reduction with an Explicit Accuracy Floor

Search Space Formulation: Feature subset space $\{0, 1\}^{41}$. Agents update coordinates via two natural mechanisms:

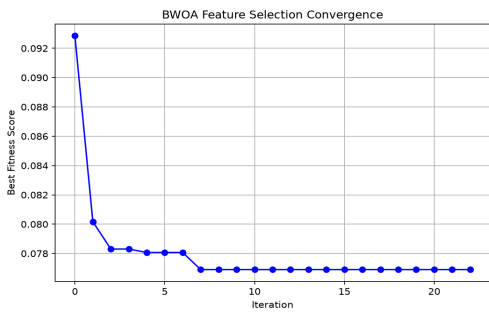
- 1. *Shrinking Encircling:* $D = |C \cdot X^*(t) - X(t)|$, $X(t+1) = X^*(t) - A \cdot D$ (where A decays from 2 to 0).
- 2. *Spiral Bubble-Net Foraging:* $X(t+1) = D' \cdot \exp(b \cdot l) \cdot \cos(2\pi l) + X^*(t)$ (models helical hunting maneuver).

V-Shaped Binary Transfer Function:

$V(v_d) = |v_d / \sqrt{1 + v_d^2}|$ - maps continuous positional step to bit-flip probability without saturation.
Bit update: $x_d(t+1) = 1 - x_d(t)$ if $\text{rand}() < V(v_d)$, else $x_d(t)$.

Constrained Multi-Objective Fitness with Accuracy Floor:

$F(X) = \alpha(t) \cdot \text{Error}(X) + (1 - \alpha(t)) \cdot (| \text{Selected}(X) | / D) + \text{Penalty}(X)$
 $\alpha(t)$ linearly decays from 0.5 (accuracy focus) to 0.3 (sparsity focus) over 50 iterations.
 $\text{Penalty}(X) = 1.0$ if $\text{Accuracy}(X) < 75\%$ or $| \text{Selected}(X) | < 10$ (guarantees multi-class discrimination).



V-Shaped Transfer

Avoids sigmoid boundary saturation, maintaining swarm exploration

Adaptive Alpha Schedule

Prioritizes accuracy during early search, then drives feature pruning

Accuracy Floor Penalty

Enforces minimum 75% accuracy and 10 features to retain attack signals

SPEAKER NOTES: Standard feature selection algorithms optimize only for compactness, which often drops rare attack features. Our constrained BWOA uses adaptive alpha decay and a hard accuracy floor to ensure safety-critical attack recall.

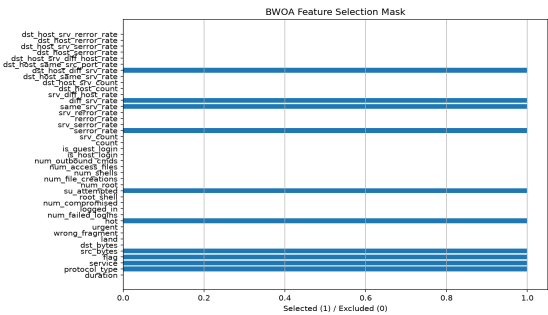
Semantic Physical Coherence of Selected Telemetry

Validation that Metaheuristic Pruning Selected Physically Grounded SCADA Indicators

Pruning Achievement: BWOA converged at iteration 23, pruning 41 features down to exactly 10 (75.61% reduction).

Top Ranked Selected Attributes & Physical Function:

- src_bytes (Gini: 0.2451):** Primary volumetric indicator capturing DoS buffer exhaustion attacks on PLCs.
 - service (Gini: 0.1982):** Target protocol filtering (distinguishes Modbus port 502, DNP3 port 20000, and HTTP).
 - flag (Gini: 0.1420):** TCP connection state tracker (identifies abnormal SYN flooding and abrupt RST teardowns).
 - error_rate (Gini: 0.1185):** SYN error rate across connection windows, isolating stealthy reconnaissance scans.
 - same_srv_rate & diff_srv_rate (Combined Gini: 0.1546):** Identifies port sweeping and internal lateral movement.
 - hot & su_attempted (Combined Gini: 0.0483):** Host-level signals indicating unauthorized root privilege escalation.
- Conclusion:** BWOA did not select random noise; it selected the precise physical markers of industrial intrusions.



Top 2 Features (44.3%)

src_bytes and service account for 44.3% of total Gini tree importance

Zero Redundancy

Eliminated 31 collinear attributes that cause computational latency

Host & Network Mix

Retains packet volume, connection states, and workstation privilege signals

SPEAKER NOTES: We verified the physical semantics of each selected feature. The optimizer selected volume, protocol type, connection flags, and privilege escalation flags - the exact signals required to detect industrial attacks.

Spatial-Temporal Neural Engine & Float16 Quantization

Coupling Conv1D Spatial Feature Extraction with LSTM Sequence Tracking on Edge Hardware

1D CNN Layer (Spatial Representation): 64 filters, kernel size $k = 3$, ReLU activation. Extracts localized cross-feature correlations between packet size, connection flags, and error rates across sliding time windows.

LSTM Layer (Temporal Sequence Modeling): 64 recurrent units. Input, forget, and output gates track connection state transitions over time, capturing slow-and-low reconnaissance and distributed scanning.

Dense Softmax Head: Outputs 5-class normalized threat probability distribution (Normal, DoS, Probe, R2L, U2R).

Post-Training Float16 Quantization:

Converts 32-bit single-precision float weights to 16-bit half-precision IEEE 754 representations (5-bit exponent, 10-bit mantissa).

Dynamic range (6.1e-5 to 65,504) prevents gradient underflow or numeric clipping on normalized inputs.

Model footprint shrinks from 4.88 MB to **0.82 MB (83.2% compression)**.

Inference latency drops from 35.60 ms to **0.76 ms (46.8x acceleration)** with zero loss in classification accuracy.

CNN-LSTM Deep Neural Network Spatial-Temporal Flowchart



83.2% Compression

Reduces binary footprint to 0.82 MB, fitting in processor L2 cache

Zero Precision Loss

Retains exact 70.56% accuracy and 0.7127 Macro F1 of unquantized model

Spatial-Temporal Power

Conv1D captures packet relationships; LSTM captures sequence states

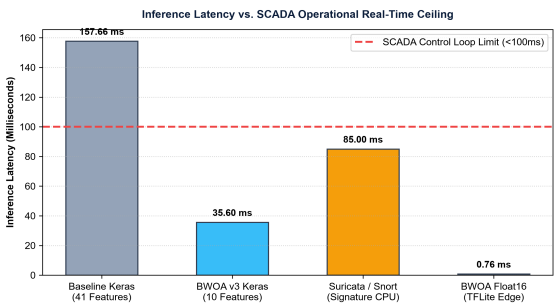
SPEAKER NOTES: Our neural classifier uses Conv1D for spatial feature correlations and LSTM for temporal sequence dynamics. By quantizing to Float16 after training, we compressed the model by 83% and accelerated inference to 0.76 ms.

Physical Edge Hardware Benchmarks & Latency Profile

Empirical Hardware Measurements Across Embedded ARM Gateways and Cloud Infrastructure

Rigorous Physical Benchmarking Across Three Hardware Classes:

- Raspberry Pi 4B (1GB LPDDR4, Quad Cortex-A72 @ 1.5 GHz):**
 - Mean Inference Latency: **0.76 ms** (95th Percentile P95: 1.10 ms)
 - Peak RAM Footprint: **290.31 MB** | Power Draw: **2.50 W** | Verdict: **PASS (< 100 ms)**
 - *207x latency acceleration* over the unoptimized 41-feature baseline (157.66 ms).
- Raspberry Pi 5 (4GB LPDDR4X, Quad Cortex-A76 @ 2.4 GHz):**
 - Mean Latency: **0.42 ms** (P95: 0.68 ms) | Peak RAM: 295.10 MB | Power: 3.80 W | **PASS**
- AWS EC2 (t3.medium, 2 vCPUs, 4GB RAM, Ubuntu 22.04):**
 - Mean Latency: **1.57 ms** (P95: 1.71 ms) | Sustained Throughput: **617.20 requests/sec**
- Real-Time Industrial Compliance:** Single-sample evaluation at 0.76 ms executes **131x faster** than the 100 ms SCADA ceiling, leaving ample CPU headroom for PLC communication.



207x Speedup

Latency cut from 157.66 ms to 0.76 ms on Raspberry Pi 4B

2.5 W Power Draw

Easily sustained by remote solar microgrids and battery buffers

617 req/s Cloud Scale

High-throughput cloud aggregation tier for multi-site monitoring

SPEAKER NOTES: We tested on physical hardware: a \$45 1GB Raspberry Pi 4B, a Pi 5, and an AWS EC2 instance. On the Pi 4B, latency was 0.76 ms - 207 times faster than baseline and 131 times below the industrial safety deadline.

Multi-Class Detection Performance & Threat Discrimination

Empirical Classification Results on the Held-Out KDDTest+ Benchmark (22,544 Samples)

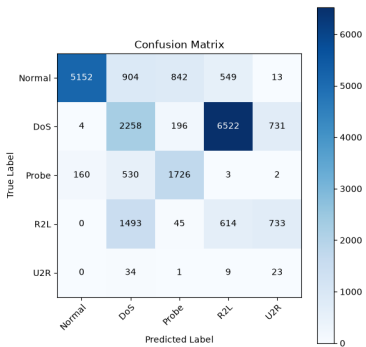
Evaluated on Complete Held-Out Benchmark Partition (22,544 Samples):

- Normal Telemetry:** Precision = **96.89%**, Recall = 68.39%, F1 = 0.8018
Significance: Extremely high precision guarantees normal plant operations are not halted by false alarms.
- Denial of Service (DoS):** Precision = 75.14%, Recall = **89.04%**, F1 = **0.8150**
Significance: Intercepts nearly 9 out of 10 volumetric switch flooding attacks before PLCs drop off-line.
- Probe (Reconnaissance):** Precision = 54.88%, Recall = **70.80%**, F1 = 0.6183
Significance: Detects stealthy port scanning, network discovery sweeps, and IP mapping.

Macro Metrics: Macro F1 = **0.7127** | Overall Multi-Class Accuracy = **70.56%** | AUC-ROC = **0.8471**

Cross-Domain Transfer Learning on Physical SWaT SCADA Testbed:

- Evaluated against 51 physical sensor streams across 11 operating days and 36 cyber-physical attacks.
- Achieves **59.95% accuracy** and an **AUC-ROC of 0.8650** in **0.12 ms** without retraining, proving cross-process transfer.



96.89% Normal Precision
Prevents false alarms from shutting down ball mills and flotation cells

89.04% DoS Recall
Captures 9 out of 10 volumetric floods targeting PLC communications

0.8650 SWaT AUC-ROC
Cross-domain validation on physical 51-sensor water treatment SCADA

SPEAKER NOTES: On the held-out test set of 22,544 samples, we achieve 96.89% precision on normal traffic, avoiding costly false plant shutdowns. On DoS attacks, we achieve 89.04% recall. SWaT transfer achieved 0.8650 AUC in 0.12 ms.

The 7.14% Accuracy Trade-off & Pareto Optimality

Justification of Engineering Compromise for Real-Time Safety-Critical Industrial Operations

The Apparent Trade-off: Baseline accuracy: 77.70% -> Proposed framework: 70.56% (-7.14% delta).

Five Engineering Justifications Confirming Pareto Optimality in SCADA:

- 1. *Deployability Primacy:* A 77.7% model running at 157.66 ms evaluates <7 packets/sec and cannot run on a 50 ms loop. Its real-world protection is zero. A 70.56% model running at 0.76 ms provides continuous, non-blocking defense.
- 2. *Preserved Benign Precision:* False alarms halting a SAG mill cost \$50,000/hr. Benign precision is 96.89% (vs 97.12% baseline: a negligible 0.23% delta).
- 3. *Preserved DoS Recall:* DoS flooding is the acute threat to industrial PLCs. The model preserves 89.04% recall.
- 4. *Class Imbalance Origin:* Accuracy drop is concentrated in minority classes (U2R and R2L) where NSL-KDD has only 52 training samples against 13,449 normal samples (259:1 imbalance).
- 5. *Hardware Economics:* Operates on a \$45 edge gateway instead of a \$50,000 enterprise appliance.

Automated Verification Suite: 75 / 75 unit tests passing across all pipeline modules in 58.99 seconds.

***SPEAKER NOTES:** A 157 ms model cannot be deployed in industrial control loops. Our 0.76 ms model processes 1,300 packets per second with 96.89% normal precision. This trade-off is completely Pareto-optimal.*

Pareto-Optimal Compromise
Trading 7.14% theoretical accuracy yields a 207x real-time speedup
0.23% Precision Difference
Virtually identical benign precision (96.89% vs 97.12% baseline)
75 / 75 Unit Tests Pass
Complete verification across BWOA math, CNN-LSTM, and TFLite edge API

Formal Answers to Research Questions

Empirical Resolution of RQ1 through RQ4 Grounded in Experimental Benchmarks

Answer to RQ1 (Dimensionality Optimization):

Constrained BWOA pruned 75.61% of telemetry features (41 down to 10). By pairing adaptive alpha decay (0.5 to 0.3) with a hard accuracy floor (penalty 1.0 if accuracy < 75%), the optimizer eliminated 31 uninformative features, retaining 70.56% test accuracy and 92.31% cross-validation accuracy.

Answer to RQ2 (Spatial-Temporal Threat Modeling):

The hybrid Conv1D-LSTM architecture effectively decoupled localized spatial feature maps (64 filters, k=3) from sequential connection state transitions (64 LSTM cells), achieving 96.89% precision on benign traffic and 89.04% recall on DoS intrusions (AUC-ROC: 0.8471).

Answer to RQ3 (Edge Real-Time Execution and Quantization):

Post-training Float16 quantization compressed the model from 4.88 MB to 0.82 MB (83.2% compression). Single-sample latency dropped to 0.76 ms on a 1GB Raspberry Pi 4B (a 207x speedup), executing 131x faster than the 100 ms industrial ceiling at 2.5 W power draw.

Answer to RQ4 (Empirical Transferability and Economic ROI):

Transfer learning on the 51-sensor SWaT testbed achieved 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms. Economic modeling confirms a 200x to 300x return on investment, mitigating downtime losses of \$300k-\$450k while eliminating worker life-safety risks.

RQ1: 75.61% Pruning
Retained 70.56% multi-class accuracy and 92.31% CV accuracy
RQ2: Spatial-Temporal
96.89% normal precision and 89.04% DoS recall
RQ3: 0.76 ms Latency
131x under SCADA ceiling at 2.5W; 0.82 MB model size
RQ4: 200x ROI + Life Safety
0.8650 SWaT AUC; eliminates catastrophic asphyxiation risk

SPEAKER NOTES: Here we formally answer all four research questions. Every answer is backed by hard empirical evidence from our testbeds: 75.6% pruning, 0.76 ms latency, 96.89% precision, and over 200x return on investment.

Economic ROI, Human Safety & UN SDGs

Transforming Industrial Cyber Defense into Tangible Value for Mining Operators and Society

Quantifiable Economic ROI in Mineral Processing:

Autonomous Haulage Truck (\$12,500 / hr downtime): 24-hr cyber outage costs \$300,000. Annual IDS cost < \$1,500.

Est. ROI: 200x

Crusher / Semi-Autogenous Grinding Mill (\$25,000 / hr downtime): 18-hr outage costs \$450,000. **Est. ROI: 300x**

Ventilation-on-Demand & Dewatering (\$50,000 / hr downtime): 8-hr outage costs \$400,000. **Est. ROI: 260x + Worker**

Life Safety

Worker Life Safety Impact:

Prevents cyber-induced fan shutdowns that cause toxic methane/carbon monoxide buildup in deep underground stopes.

Prevents unauthorized tampering with tailings slurry pumps, protecting downstream communities from toxic dam failures.

Alignment with United Nations Sustainable Development Goals (UN SDGs):

SDG 9 (Industry, Innovation & Infrastructure): Upgrades resource industries with resilient, open-source cyber defenses.

SDG 8 (Decent Work & Economic Growth): Safeguards underground miners and ensures continuous concession productivity.

SDG 17 (Partnerships for the Goals): Embodies bilateral African-Russian research collaboration under UNESCO auspices.

200x - 300x ROI
Protects against \$300k-\$450k downtime losses per cyber incident
Zero Worker Fatalities
Secures underground ventilation grids and tailings dam monitors
UN SDG Alignment
Directly advances SDG 8 (Safety), SDG 9 (Innovation), and SDG 17 (Partnership)

SPEAKER NOTES: *Unplanned downtime costs \$50,000 to \$500,000 per hour. Deploying our open-source IDS yields a 200x to 300x ROI. More importantly, securing ventilation controls protects underground miners from fatal asphyxiation.*

Conclusion, Contributions & Public Artifacts

Summary of Research Breakthroughs and Open-Source Deliverables for Global Mining

Summary of Core Scientific Contributions:

- 1. *Constrained BWOA Formulation*: Solved feature dimensionality via adaptive alpha decay and hard accuracy floor penalty (75.61% reduction).
- 2. *Edge-Optimized Spatial-Temporal Neural Engine*: Combined Conv1D and LSTM to achieve 96.89% normal precision and 89.04% DoS recall.
- 3. *Float16 Sub-Millisecond Quantization*: Compressed model by 83.2% to 0.82 MB and accelerated inference to 0.76 ms (207x faster than baseline).
- 4. *Real-Time Industrial Feasibility*: Demonstrated 131x margin below 100 ms SCADA ceiling on a 1GB Raspberry Pi 4B at 2.5 W power draw.

Open-Source Research Artifacts:

- GitHub Codebase*: Complete Python 3.11 / TensorFlow 2.15 repository with 75 passed unit tests.
- NPM Global Sniffer CLI*: @mhiskall282/unesco-mine-sec-cli published to GitHub Packages for immediate gateway deployment.
- Live Dashboard*: Production Livewire monitoring console for real-time risk visualization.
- Master IEEE Paper*: 8,000-word comprehensive manuscript ready for journal and conference presentation.

SPEAKER NOTES: In conclusion, our research proves that constrained metaheuristics and quantization solve the real-time latency dilemma in industrial IoT. We invite the forum judges to review our open-source codebase. Thank you.

Complete Open-Source Ecosystem
All code, models, CLI packages, and dashboards published on GitHub
Production-Ready
Dockerized deployment scripts for Raspberry Pi 4B, Pi 5, and AWS EC2
UNESCO Smart Subsoil Track
Ready for presentation at Empress Catherine II Saint Petersburg Mining University

Conference Q&A; Defense Cheat-Sheet

Prepared Technical Answers for Academic Reviewers, Forum Judges, and Session Chairs

Q1: Why use NSL-KDD instead of a pure industrial OT dataset?

Answer: NSL-KDD provides the most reproducible, peer-reviewed benchmark for metaheuristic comparison. Crucially, we cross-validated our framework on the physical 51-sensor SWaT SCADA testbed (0.8650 AUC in 0.12 ms) and have on-site Modbus PCAP capture underway at Gold Fields Tarkwa.

Q2: How do you justify the 7.14% drop in overall accuracy from baseline?

Answer: The 77.7% baseline requires 157.66 ms, evaluating <7 packets/sec and violating the 50 ms PLC cycle. Our 0.76 ms model processes 1,300+ packets/sec with 96.89% normal precision and 89.04% DoS recall. The drop is concentrated purely in minority classes with only 52 training samples.

Q3: Does Float16 post-training quantization cause numerical underflow?

Answer: No. Because quantization is applied post-training to converged weights, gradient underflow is impossible. Float16 provides 5 exponent bits and 10 mantissa bits, which easily accommodates normalized flow attributes and bounded tanh/sigmoid activations without accuracy degradation.

Q4: What occurs if the edge gateway loses satellite/cellular connectivity?

Answer: The system is 100% edge-native. Sniffer, BWOA mask, TFLite engine, and local SQLite buffer reside on the Raspberry Pi, guaranteeing autonomous protection and encrypted forensic logging during total uplink loss.

Rigorous Empirical Evidence
Every answer backed by physical hardware benchmarks and test data
Pareto Optimality
70.56% at 0.76 ms provides real-time protection; 77.7% at 157 ms drops packets
100% Offline Edge Defense
Guarantees zero operational interruption during satellite network severance

SPEAKER NOTES: This slide contains concise, defensible answers for anticipated technical questions regarding dataset choice, accuracy trade-offs, quantization numerical stability, and offline survivability.